

AIM

Implementation of DHCP, APIPA and analysis of FTP packets using Cisco Packet Tracer.

OBJECTIVES

1. Understanding the use of Dynamic Host Control Protocol (DHCP) and Automatic Private IP Addressing (APIPA).
2. An overview on message communication between two end hosts using FTP packets.
3. Configuring a server device as DHCP server in a network that dynamically assigns IP addresses to client PCs and verifying of IP address using APIPA the connectivity in the network using ping command.
4. Demonstrating the generation and verification of IP Address using APIPA, when a host fails to obtain an IP addressing from a DHCP server.
5. Configuring a router (in the absence of a server) in a network that implements DHCP protocol for allocation of IP address to hosts dynamically and verifying the working of DHCP.
6. Configuring a client-server network using FTP server and analysing the message communication (packets) exchanged between the client and server.

THEORY

DHCP: DHCP is a network service that automatically assigns an IP address to the devices (like PCs, printers) when they connect to network. Instead of manually setting an IP address, DHCP does it automatically.

APIPA: APIPA is a backup system used when a device cannot get an IP address from the DHCP server. If no DHCP server is available, the device assigns itself an IP address from a specified range. This allows local communication with nearby devices but no access to the internet.

FTP: Used to transfer files, uploading/downloading files b/w two devices. It requires login, then files are uploaded, downloaded or viewed using respective commands.

How it works?

- Setting up connection
- login
- upload/download files
- Close the connection

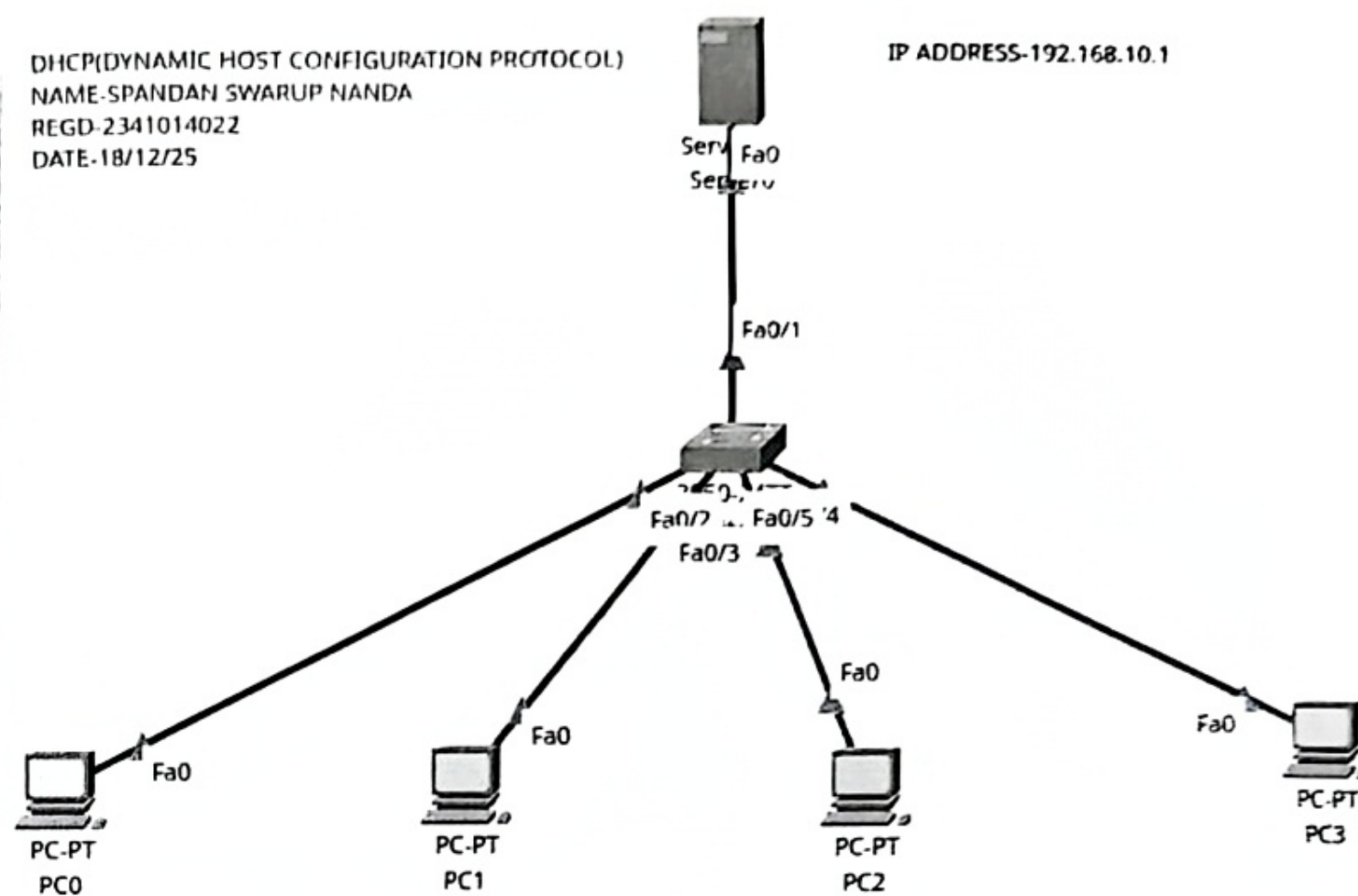
OBSERVATION

(i) DHCP

Fig. A star topology n/w design implementing a DHCP server to manage IP allocation for multiple client PCs.
↳ This diagram clearly establishes the physical connections where the central switch links the DHCP server (192.168.10.1) to the end devices.

DHCP(DYNAMIC HOST CONFIGURATION PROTOCOL)
NAME-SPANDAN SWARUP NANDA
REGD-2341014022
DATE-18/12/25

IP ADDRESS-192.168.10.1



PC0

Physical Config Desktop Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C9FF:FE10:B1

Default Gateway

DNS Server

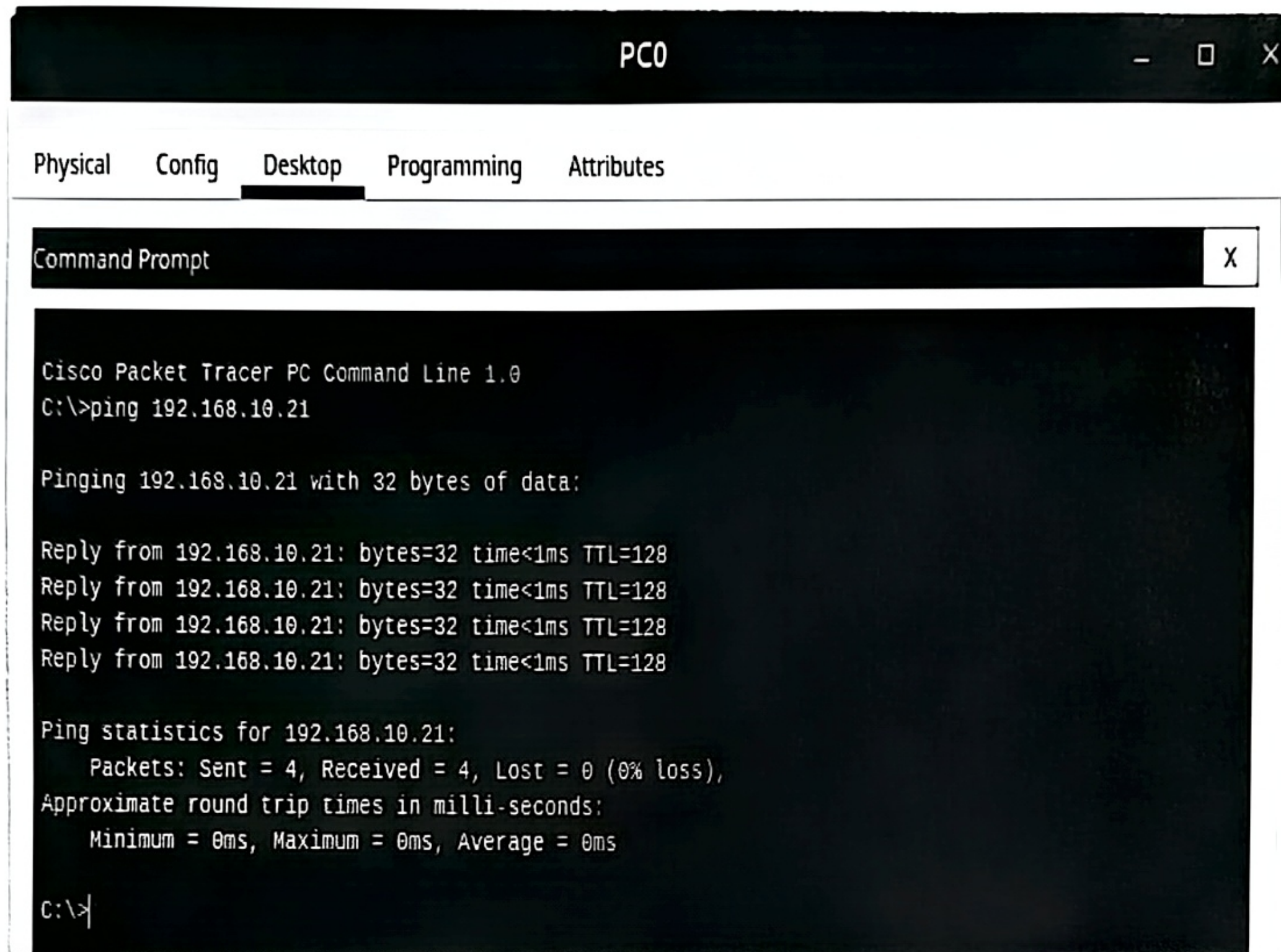
802.1X

☐ Use 802.1X Security

Authentication

Username

Fig. Client PC successfully requesting and receiving dynamic n/w configuration details via DHCP.
↳ This validates that the DHCP service is functional, as the client automatically obtained the IP address 192.168.10.20 without manual entry.



The screenshot shows a window titled 'PC0' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Reply from 192.168.10.21: bytes=32 time<1ms TTL=128
Reply from 192.168.10.21: bytes=32 time<1ms TTL=128
Reply from 192.168.10.21: bytes=32 time<1ms TTL=128
Reply from 192.168.10.21: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.21:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig. Successful connectivity verification between network devices using the ping command with zero packet loss.

↳ The output confirms that the network is stable and the client can communicate with the target IP address with a latency of less than 1 ms.

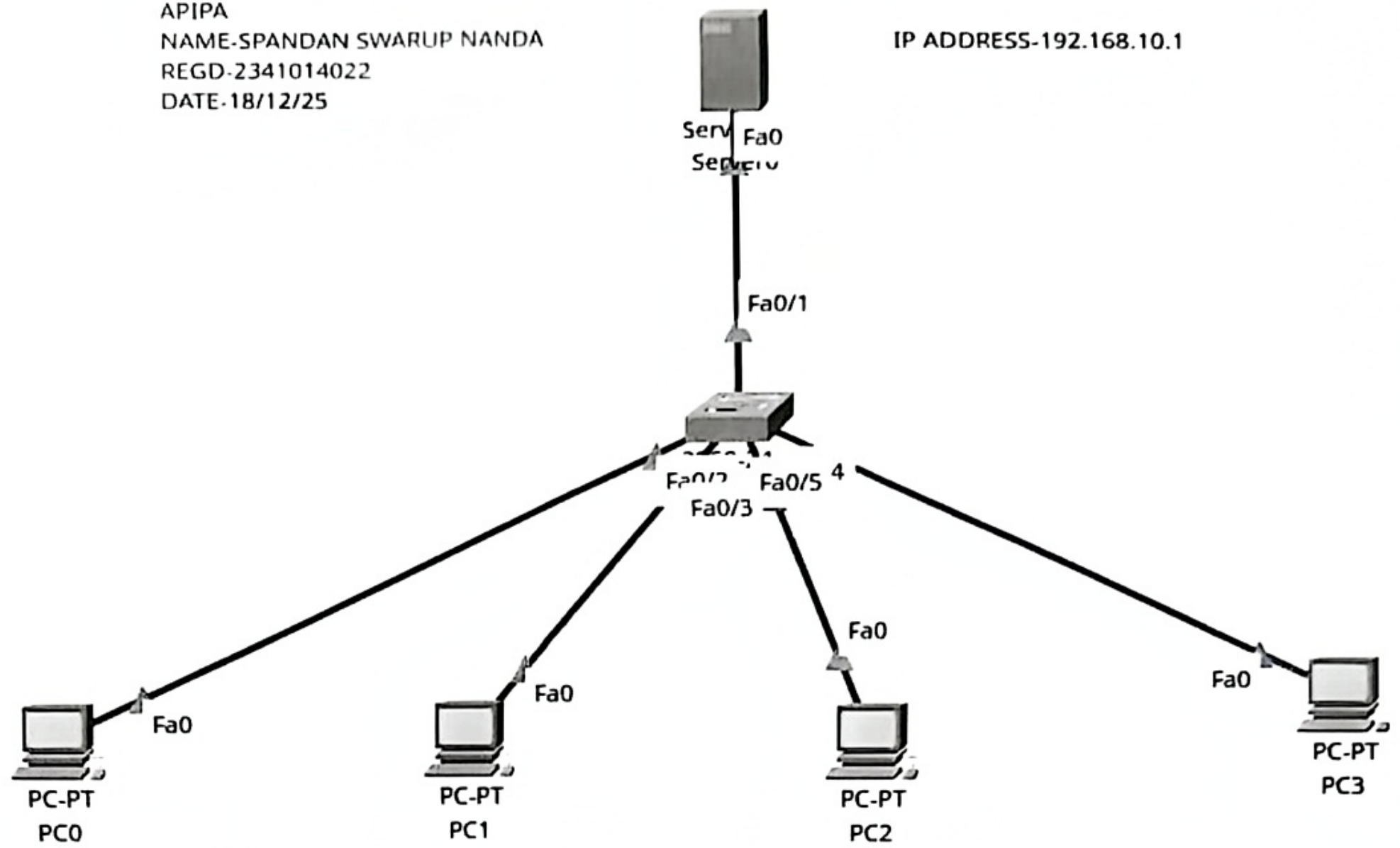
P.T.O.

2) APIPA

Fig. Demonstrating Apiipa fallback protocol in Cisco Packet Tracer
 ↳ Clear example of the client self-assigning a 169.254.0.177 address because the DHCP server is unreachable or misconfigured.

APIPA
 NAME-SPANDAN SWARUP NANDA
 REGD-2341014022
 DATE-18/12/25

IP ADDRESS-192.168.10.1



PC0

Physical
Config
Desktop
Programming
Attributes

IP Configuration
X

Interface
FastEthernet0

IP Configuration

☒ DHCP

☐ Static

DHCP failed. APIPA is being used.

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

IPv6 Configuration

☐ Automatic

☒ Static

/

IPv6 Address

Link Local Address

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

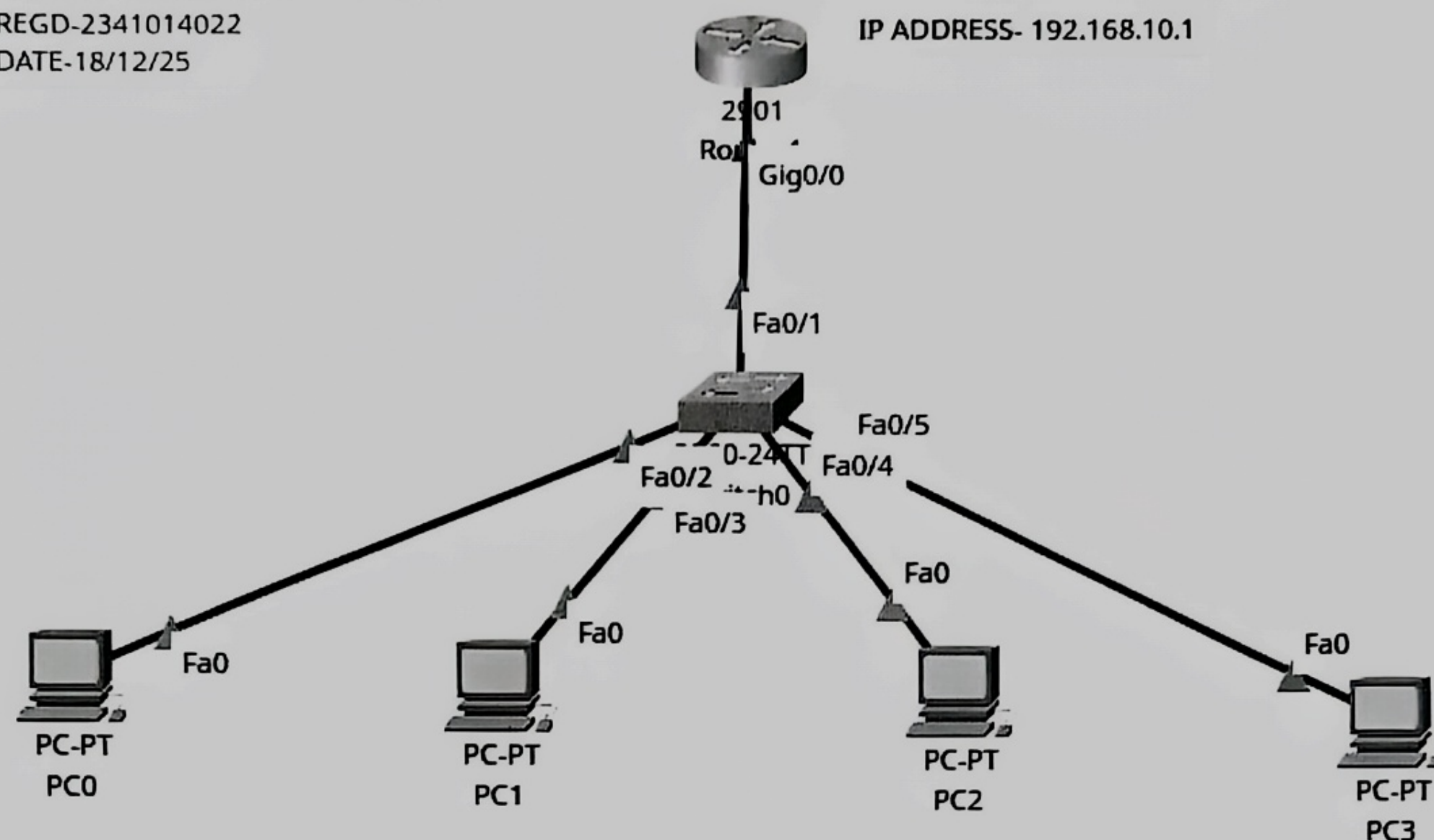
Username

Name: _____

Regd. Number: _____

(3) CISCO AS A SERVER ROUTER

CISCO ROUTER AS SERVER
NAME-SPANDAN SWARUP NANDA
REGD-2341014022
DATE-18/12/25



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

Reply from 192.168.10.12: bytes=32 time=7ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>
```

Fig. Cisco Packet Tracer simulation demonstrating a "Router as Server" topology with successful connectivity verification via ping.

↳ The successful echo replies with 0% packet loss confirms that the router is correctly configured and stable communication exists across the network.

Fig. Verification of DHCP auto configuration on PC0 after setting up the "LAN_POOL" DHCP scope on Router1 in CPT. By the router's CLI confirms the DHCP pool configuration and included addresses, which is successfully validated by PC0 automatically acquiring the first available IP address, 192.168.10.11.

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

Router(config-if)#interfcae gig0/0
      ^
% Invalid input detected at '^' marker.

Router(config-if)#interface gig0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#exit
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.10
Router(config)#ip dhcp pool LAN_POOL
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
```

PC0

Physical Config Desktop Programming Attributes

IP Configuration X

Interface	FastEthernet0
-----------	---------------

IP Configuration

☒ DHCP

☐ Static

IPv4 Address

Subnet Mask

Default Gateway

DNS Server

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address

Default Gateway

DNS Server

4) CLIENT-SERVER NETWORK USING FTP

CLIENT-SERVER NETWORK USING FTP
NAME-SPANDAN SWARUP NANDA
REGD-2341014022
SEC-23412K1
DATE-18/12/25

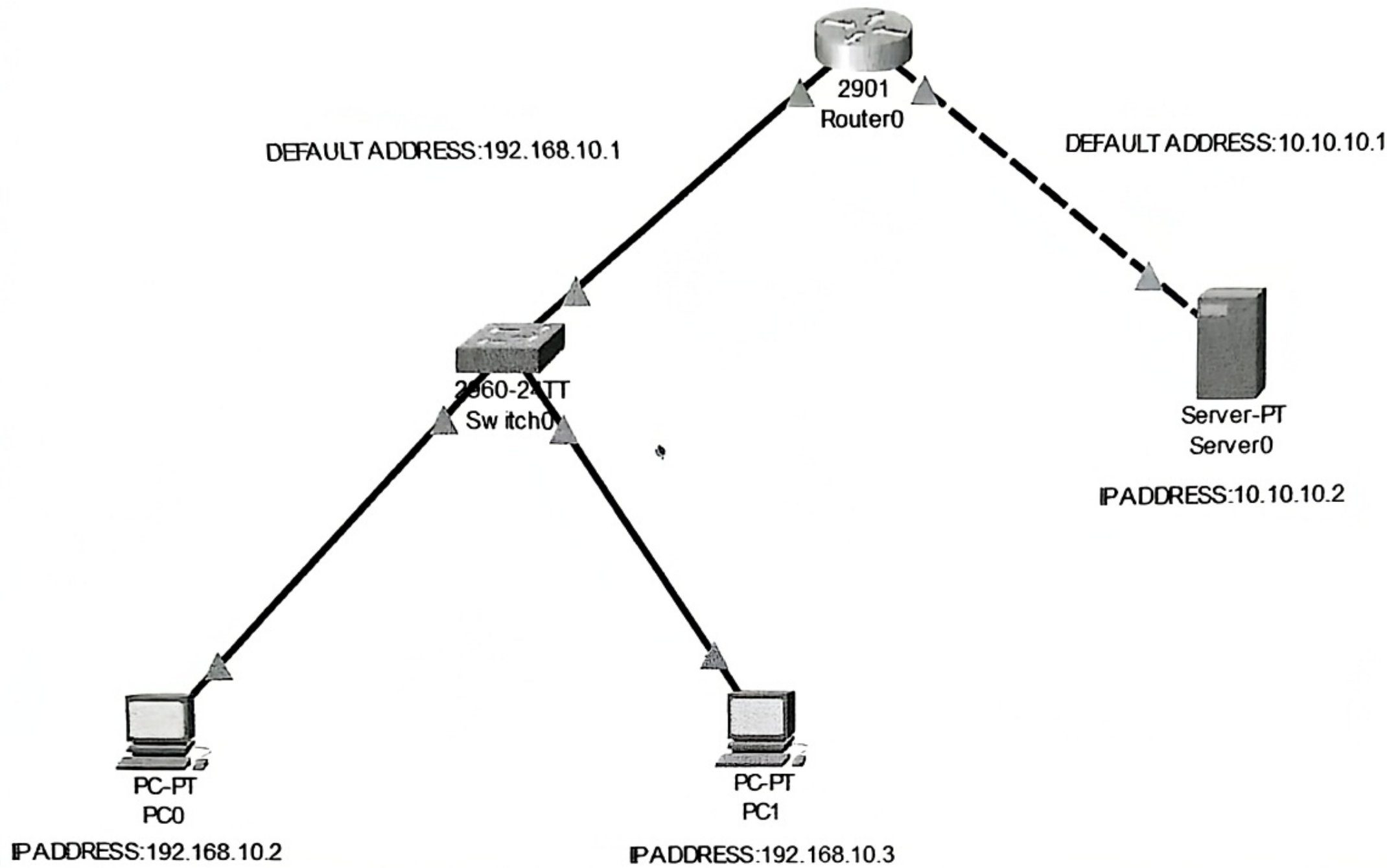


Fig. Finalized network topology connecting the client subnets to the FTP server via a central router.

↳ Visualizes the physical links and IP addressing required for the 192.168.10.x and 10.10.10.x networks to communicate.



Fig. Creating a sample text file on the Server0 desktop to serve as test data.

↳ This prepares the specific file payload that the clients will request and download during the FTP verification step.

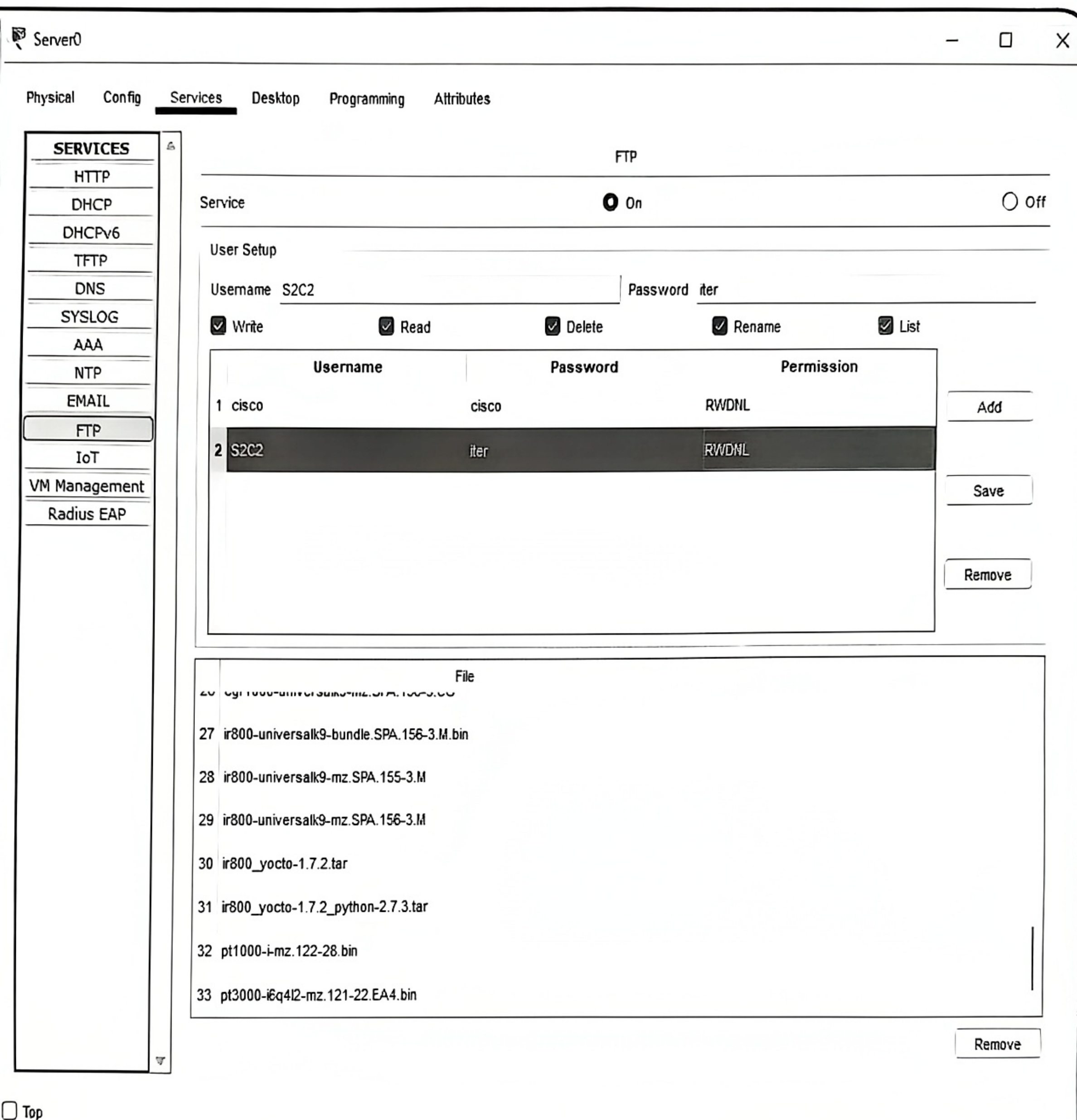
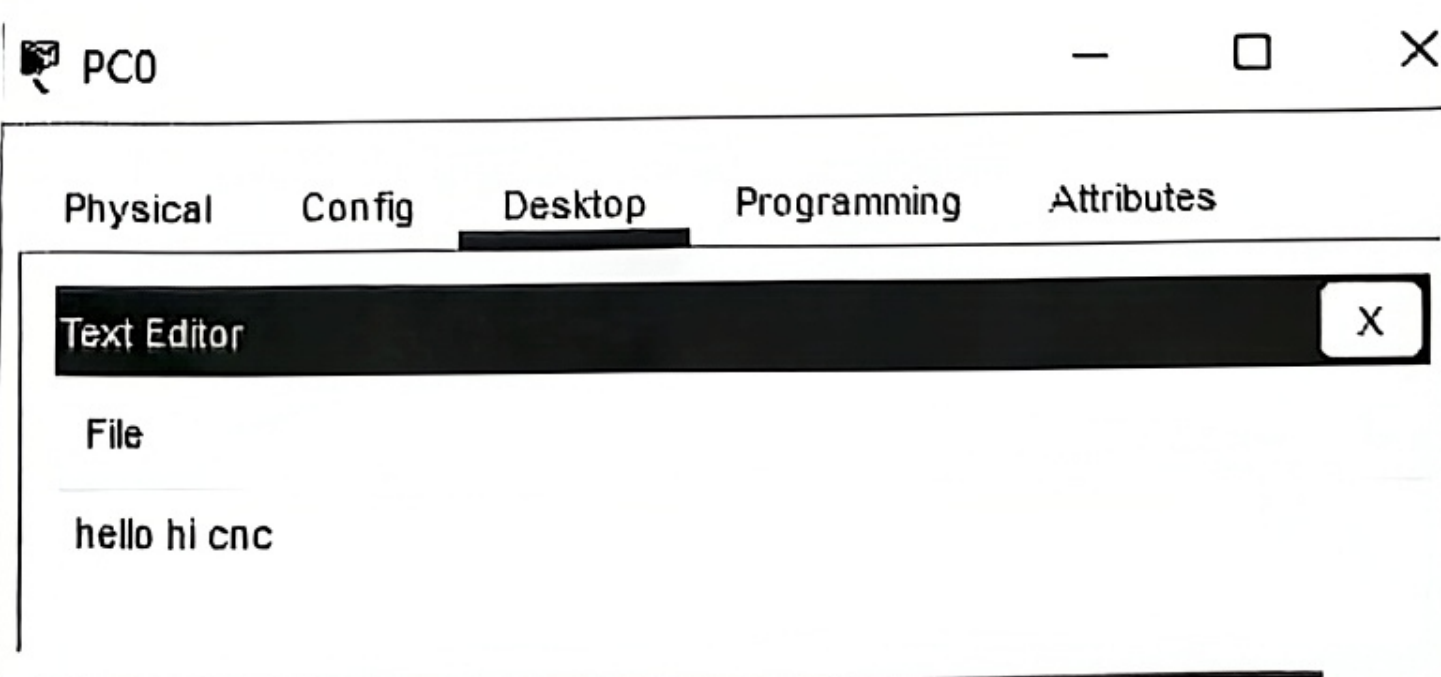


Fig. Configuring the FTP service on Server0 by adding a user "S2C2" with full Read/Write permissions.
This step secures the server by requiring authentication (username/password) before allowing file access.

Fig. Verifying the successful download of the test file on the client machine (PC0) by opening the file on the client side confirms that the entire FTP cycle - connection, authentication and transfer - worked correctly.



Command Prompt

X

Fig. Uploading the file from PC to the server using the put test.txt command.
 ↳ The "Transfer complete" message confirms that the client successfully wrote data to the server's directory.

Cisco Packet Tracer PC Command Line 1.0

C:\>dir

Volume in drive C has no label.
 Volume Serial Number is 5E12-4AF3
 Directory of C:\

1/1/1970 5:30 PM 26 sampleFile.txt
 26 bytes 1 File(s)

C:\>ftp 10.10.10.2

Trying to connect...10.10.10.2

Connected to 10.10.10.2

220- Welcome to PT Ftp server

Username:S2C2

331- Username ok, need password

Password:

230- Logged in

(passive mode On)

ftp>dir

Listing /ftp directory from 10.10.10.2:

0	:	asa842-k8.bin	5571584
1	:	asa923-k8.bin	30468096
2	:	cl841-advipservicesk9-mz.124-15.T1.bin	33591768
3	:	cl841-ipbase-mz.123-14.T7.bin	13832032
4	:	cl841-ipbasek9-mz.124-12.bin	16599160
5	:	cl900-universalk9-mz.SPA.155-3.M4a.bin	33591768
6	:	c2600-advipservicesk9-mz.124-15.T1.bin	33591768
7	:	c2600-i-mz.122-28.bin	5571584
8	:	c2600-ipbasek9-mz.124-8.bin	13169700
9	:	c2800nm-advipservicesk9-mz.124-15.T1.bin	50938004
10	:	c2800nm-advipservicesk9-mz.151-4.M4.bin	33591768
11	:	c2800nm-ipbase-mz.123-14.T7.bin	5571584
12	:	c2800nm-ipbasek9-mz.124-8.bin	15522644
13	:	c2900-universalk9-mz.SPA.155-3.M4a.bin	33591768
14	:	c2950-i6q412-mz.121-22.EA4.bin	3058048
15	:	c2950-i6q412-mz.121-22.EA8.bin	3117390
16	:	c2960-lanbase-mz.122-25.FX.bin	4414921
17	:	c2960-lanbase-mz.122-25.SEE1.bin	4670455
18	:	c2960-lanbasek9-mz.150-2.SE4.bin	4670455
19	:	c3560-advipservicesk9-mz.122-37.SEE1.bin	8662192
20	:	c3560-advipservicesk9-mz.122-46.SE.bin	10713279
21	:	c800-universalk9-mz.SPA.152-4.M4.bin	33591768
22	:	c800-universalk9-mz.SPA.154-3.M6a.bin	83029236
23	:	cat3k_caa-universalk9.16.03.02.SPA.bin	505532849
24	:	cgr1000-universalk9-mz.SPA.154-2.CG	159487552
25	:	cgr1000-universalk9-mz.SPA.156-3.CG	184530138
26	:	ir800-universalk9-bundle.SPA.156-3.M.bin	160968869
27	:	ir800-universalk9-mz.SPA.155-3.M	61750062
28	:	ir800-universalk9-mz.SPA.156-3.M	63753767
29	:	ir800_yocto-1.7.2.tar	2877440
30	:	ir800_yocto-1.7.2_python-2.7.3.tar	6912000
31	:	pt1000-i-mz.122-28.bin	5571584
32	:	pt3000-i6q412-mz.121-22.EA4.bin	3117390
33	:	test.txt	12

ftp>get test.txt

Reading file test.txt from 10.10.10.2:

File transfer in progress...

[Transfer complete - 12 bytes]

12 bytes copied in 0 secs

ftp>exit

Invalid or non supported command.

ftp>quit

221- Service closing control connection.

C:\>dir

Volume in drive C has no label.
 Volume Serial Number is 5E12-4AF3
 Directory of C:\

1/1/1970 5:30 PM 26 sampleFile.txt
 1/1/1970 5:30 PM 12 test.txt
 38 bytes 2 File(s)

C:\>

PC0

Fig. Downloading the file to a second client (PC1) using the get tent.txt command
↳ Completes the sharing cycle by retrieving the file from the central server to a different machine.

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>dir

Volume in drive C has no label.

Volume Serial Number is 5E12-4AF3

Directory of C:\

1/1/1970	5:30 PM	26	sampleFile.txt
1/1/1970	5:30 PM	12	test.txt
		38 bytes	2 File(s)

C:\>ftp 10.10.10.2

Trying to connect...10.10.10.2

Connected to 10.10.10.2

220- Welcome to PT Ftp server

Username:S2C2

331- Username ok, need password

Password:

230- Logged in

(passive mode On)

ftp>dir

Listing /ftp directory from 10.10.10.2:

0	: asa842-k8.bin	5571584
1	: asa923-k8.bin	30468096
2	: cl841-advipservicesk9-mz.124-15.T1.bin	33591768
3	: cl841-ipbase-mz.123-14.T7.bin	13832032
4	: cl841-ipbasek9-mz.124-12.bin	16599160
5	: cl900-universalk9-mz.SPA.155-3.M4a.bin	33591768
6	: c2600-advipservicesk9-mz.124-15.T1.bin	33591768
7	: c2600-i-mz.122-28.bin	5571584
8	: c2600-ipbasek9-mz.124-8.bin	13169700
9	: c2800nm-advipservicesk9-mz.124-15.T1.bin	50938004
10	: c2800nm-advipservicesk9-mz.151-4.M4.bin	33591768
11	: c2800nm-ipbase-mz.123-14.T7.bin	5571584
12	: c2800nm-ipbasek9-mz.124-8.bin	15522644
13	: c2900-universalk9-mz.SPA.155-3.M4a.bin	33591768
14	: c2950-i6q412-mz.121-22.EA4.bin	3058048
15	: c2950-i6q412-mz.121-22.EA8.bin	3117390
16	: c2960-lanbase-mz.122-25.FX.bin	4414921
17	: c2960-lanbase-mz.122-25.SEE1.bin	4670455
18	: c2960-lanbasek9-mz.150-2.SE4.bin	4670455
19	: c3560-advipservicesk9-mz.122-37.SEE1.bin	8662192
20	: c3560-advipservicesk9-mz.122-46.SE.bin	10713279
21	: c800-universalk9-mz.SPA.152-4.M4.bin	33591768
22	: c800-universalk9-mz.SPA.154-3.M6a.bin	83029236
23	: cat3k_caa-universalk9.16.03.02.SPA.bin	50553284
24	: cgr1000-universalk9-mz.SPA.154-2.CG	15948755
25	: cgr1000-universalk9-mz.SPA.156-3.CG	18453013
26	: ir800-universalk9-bundle.SPA.156-3.M.bin	16096886
27	: ir800-universalk9-mz.SPA.155-3.M	61750062
28	: ir800-universalk9-mz.SPA.156-3.M	63753767
29	: ir800_yocto-1.7.2.tar	2877440
30	: ir800_yocto-1.7.2_python-2.7.3.tar	6912000
31	: pt1000-i-mz.122-28.bin	5571584
32	: pt3000-i6q412-mz.121-22.EA4.bin	3117390

ftp>put test.txt

Writing file test.txt to 10.10.10.2:

File transfer in progress...

[Transfer complete - 12 bytes]

12 bytes copied in 0.154 secs (77 bytes/sec)

Name: _____

ANALYSIS

DHCP Using Server

- Server successfully assigned IP addresses dynamically to client PCs.
- Manual IP configuration was eliminated.
- Ping verification confirmed proper connectivity.

APIPA Addressing

- When DHCP was unavailable, hosts auto assigned APIPA addresses (169.254.x.x)
- Limited local communication was possible.
- No gateway or internet access was available.

DHCP Using Router

- Router functioned as a DHCP server and assigned IP addresses to clients.
- DHCP bindings confirmed successful allocation.
- Suitable for small networks without dedicated servers.

FTP Client - Server Communication

- FTP server enabled file upload and download between client and server.
- TCP ensured reliable data transfer.
- Proper application - layer communication was observed.

CONCLUSION

This experiment successfully demonstrated the working of DHCP, APIPA, and FTP using Cisco Packet Tracer. Dynamic IP address allocation using both server-based and router based DHCP was verified, reducing manual configuration. The APIPA mechanism was observed when DHCP was unavailable, ensuring limited local communication. Additionally, FTP client-server communication was established and analyzed, confirming reliable file transfer betⁿ hosts. Overall, the experiment achieved all the stated objectives and strengthened understanding of dynamic addressing and application-layer communication.

EXERCISES

1. What is DHCP snooping? What are the main advantages of using DHCP in a network?
2. Set up a network with a router and two PCs using Cisco Packet Tracer. Configure DHCP on the router with the following settings:
 - (a) Network Address: 192.168.100.0/24
 - (b) DHCP Pool: Start IP: 192.168.100.20, End IP: 192.168.100.40
 - (c) Default Gateway: 192.168.100.1
 - (d) DNS Server: 8.8.8.8

(3) State the use of APIPA highlighting its advantages. What is the range of IP addresses for APIPA? Write the APIPA address generated for your device in this experiment.

(4) Mention true/false.

- (a) FTP uses two TCP connections.
- (b) FTP sends exactly one file over the data connection.
- (c) FTP server is stateless.
- (d) FTP is a secure protocol.

(1) DHCP Snooping is security feature used in networks to prevent unauthorized DHCP servers from assigning IP addresses.

Advantages:

- IP addresses are getting automatically.
- It ensures efficient use of IP addresses.

(2) DHCP Configuration on Router:-

- The Router interface was configured with IP address 192.168.100.1/24 and enabled.
- Static IP address were excluded from DHCP allocation.
- A DHCP pool was created for network 192.168.100.0/24.
- IP range 192.168.100.20 - 192.168.100.40 was assigned to clients.
- Default gateway 192.168.100.1 and DNS 8.8.8.8 were configured.
- Client PCs were set to DHCP mode.
- PCs successfully received IP addresses dynamically, verifying DHCP operation.

(3) It is used when a device cannot get an IP address from a DHCP server. It assigns an IP address automatically enabling devices to communicate within the same local network.

Advantages of APIPA -

- no need of manual configuration of IP address
- provides backup if DHCP fails.

(4)(a) True

(b) False

(c) False

(d) False

_____ X _____